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Mapping product sustainability assessment needs across the product development process to form coherent methodological support

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Abstract: Implementing sustainable product development requires assessing sustainability performance throughout the product development process. However, companies face a wide range of assessment approaches and lack guidance on how to select and combine them effectively. Research increasingly calls for coherent methodological support, or a methods ecosystem, to enable consistent adoption. This paper addresses this need by combining empirical insights on capability needs related to product sustainability performance assessment with a synthesis of literature on product development phases, decisions, and tasks. A conceptual model is proposed to show how assessment tasks and capability needs change across development stages and to clarify when effectiveness- and efficiency-oriented approaches are required. The conceptual model is intended to advance research on the creation of coherent methodological support for product sustainability performance assessment by identifying characteristics of supports needed in different stages.

Keywords: sustainable product development, product sustainability performance, design process, design for sustainability.

1 Introduction

Product development and manufacturing companies play a key role in society's transition towards sustainability (Bengtsson et al., 2018; Ceschin and Gaziulusoy, 2016). Decisions made in the early stages of product development have significant influence on the final lifecycle impact of the product (Albers et al., 2019; McAloone and Tan, 2005). Sustainable product development (SPD) is an approach that has emerged to support the integration of strategic sustainability considerations from the beginning of the product development process (PDP) (Hallstedt and Isaksson, 2017). Implementing SPD in practice demands the capability to assess, monitor and evaluate product sustainability performance (Kravchenko et al., 2019; Parolin et al., 2024; Villamil et al., 2022), which is essential for comparing design alternatives and informing decision-making across different stages of product development (Kalish et al., 2018; López-Mesa and Bylund, 2011; Mallalieu et al., 2024).

However, assessing product sustainability performance is often time-intensive and constrained by challenges related to data availability and uncertainty management, particularly in the early stages of product development, where design freedom is high but information is scarce (Hallstedt, 2017; Schöggl et al., 2017). Furthermore, companies are confronted with an overwhelming variety of approaches to assess product sustainability performance (Faludi et al., 2020; Lövdahl et al., 2024). These approaches differ substantially in their underlying assumptions, the sustainability dimensions and issues they prioritize, the degree of qualitative or quantitative data they require, and the effort needed to conduct assessments (Ahmad et al., 2018; Gaziulusoy, 2015). The proliferation of these diverse approaches has resulted in a fragmented landscape in which it is difficult to understand the role and contribution of each approach, as well as how different approaches can be combined to support the integration of strategic sustainability considerations. Therefore, product sustainability performance assessment practice is often characterized by a collection of isolated approaches, lacking the systematic integration needed to support practitioners across different stages of the PDP. This fragmentation highlights the need for a methods ecosystem, in which complementary approaches are structured to be connected rather than applied independently, supporting coherent methodological support for decision-making (Faludi et al., 2020; Gericke et al., 2020).

In response, research has approached this challenge from two perspectives: either by proposing process-oriented models, that address the different needs across the PDP (Brones et al., 2017; Diaz et al., 2021; Peace et al., 2018) or by exploring how existing tools and methods can be classified and clustered according to their purposes and characteristics, also discussing their benefits and limitations (Ahmad et al., 2018; Chebaeva et al., 2021; Faludi, 2017). However, there is limited research on how these approaches can be combined, sequenced, or navigated to provide coherent methodological support that enables companies to systematically implement product sustainability performance assessment.

This paper addresses the following research question: *how can a coherent methodological support for product sustainability performance assessment be structured to align with company capability needs and specific product development stages?*

By addressing this question, the contribution of this study is twofold. First, it synthesizes existing perspectives to explain how assessment needs and characteristics evolve across product development stages and links these dynamics to empirically identified capability needs. Second, it proposes a conceptual model that shows how sustainability performance

assessment activities evolve throughout the PDP and specifies characteristics of methodological support required, as well as their interrelationships.

2 Product sustainability performance across the product development process

As sustainability is increasingly understood as a systems property, products alone cannot be deemed sustainable or unsustainable (Gaziulusoy & Brezet, 2015). Product sustainability performance should therefore be understood from a socio-ecological systems perspective, where products' lifecycles are assessed in relation to their influence on larger social and ecological dynamics (Broman and Robèrt, 2017; Ceschin and Gaziulusoy, 2016). While many conventional sustainability assessment approaches evaluate performance using a relative point of reference (e.g., previous products or market benchmarks), absolute approaches evaluate whether impacts remain within ecological or social limits necessary to maintain system integrity. Adopting absolute reference points supports long-term strategic decision-making by clarifying whether performance is "good enough" in relation to science-based boundaries rather than merely "less bad" (Bjørn et al., 2020; Broman and Robèrt, 2017; Hauschild et al., 2020). The concept of product sustainability performance, should thus both serve as a normative function that establishes a shared vision of what a product should do to contribute to systemic sustainability, and enable comparative product sustainability performance assessments throughout the PDP (Gaziulusoy, 2015; Gebler et al., 2022; Sala et al., 2015).

Product sustainability performance should, therefore, be assessed from two complementary performance lenses: effectiveness and efficiency. An effectiveness lens applied in the early stages of product development supports decision-making by ensuring that the solutions lead in the right direction, while an efficiency lens is necessary in the later stages to ensure that solutions do so in the best way (Zimek and Baumgartner, 2024). Therefore, a balance between both lenses is necessary. Previous research has shown that most methods, tools and product sustainability related regulations have a focus on efficiency, limiting the potential for systemic transformation (Ahmad et al., 2018; Bengtsson et al., 2018; Bjørn et al., 2020; Zimek and Baumgartner, 2024).

Effectiveness and efficiency lenses support different types of decisions and therefore need to be aligned with the structure of the PDP and with the tasks that most strongly influence product sustainability performance, as identified by Vilochani et al. (2025). Product development is an iterative process that serves as a primary mechanism for translating corporate strategy into market offerings, with the overarching objective of developing products that sustain or increase competitive position (Buijs, 2003; Ulrich and Eppinger, 2012). Product planning begins with an understanding of corporate strategy and external and internal contexts, which informs the formulation of product strategies and innovation priorities (Buijs, 2003; Khurana and Rosenthal, 1998). For companies pursuing SPD, product strategy development additionally requires the integration of sustainability trends, corporate sustainability objectives, and the identification of sustainability-related risks and opportunities (Diaz et al., 2021; Schulte et al., 2020; Vilochani et al., 2025).

Translating corporate strategy into product and innovation strategies typically involves cross-functional and multi-stakeholder decision processes, such as workshops (Khurana and Rosenthal, 1998; Phaal et al., 2004; Villamil and Hallstedt, 2021). Such alignment supports the incorporation of sustainability into product visions, strategies, and roadmaps at a strategic level (Nappi et al., 2024; Vilochani et al., 2025). Shared vision-building is further recognized as a prerequisite for strategic decision-making using backcasting approaches in sustainability-oriented innovation and in this context should be guided by a clear understanding of what product sustainability means in each case (Broman and Robèrt, 2017; Kishita et al., 2024). Because decisions at this stage are exploratory and not yet based on defined product specifications, decision-support tools primarily facilitate opportunity exploration, awareness-building, and the convergence of sustainability objectives, thereby supporting strategic refinement of product development (Rotondo et al., 2025). In practice, this can involve the development of product sustainability criteria (e.g., Watz and Hallstedt, 2020). The sustainability criteria enable the comparison of alternative concepts in relation to their expected performance in subsequent phases of the PDP. To support systematic evaluation, the criteria are subsequently translated into measurable indicators that allow semi-quantitative and quantitative assessment as design definitions become more detailed (Hallstedt et al., 2023).

Once product strategy is established, the process moves into product development activities, which can be broadly structured into two main stages: concept development and product realization (Krishnan and Ulrich, 2001). Concept development encompasses activities such as identifying customer needs and defining product requirements (Krishnan and Ulrich, 2001). This stage is characterized by high uncertainty, limited data availability, and exploratory decision-making, as fundamental product directions are still being defined (Hallstedt, 2017; Schöggl et al., 2017). Early-stage decision support includes methods for sustainability requirement specification, self-assessment of organizational readiness, reuse of prior sustainability information (e.g., retrospective LCA), and capturing the voice of customers regarding sustainability expectations (Ahmad et al., 2018; Diaz et al., 2021).

During concept development, sustainability performance requirements based on sustainability criteria from the product strategy can be used to guide design decisions. These sustainability performance requirements are, thus, ideally sourced from the strategic level, ensuring traceability and strategic relevancy in trade-offs with other product requirements

(Mendoza et al., 2017; Nappi et al., 2024). This translation enables subsequent quantitative sustainability assessments as the process evolves to product realization, which includes engineering design, material and process selection, prototyping, testing, supply-chain configuration, and preparation for production and market launch. In these phases, uncertainty decreases and data availability increases as design definitions become more detailed. Accordingly, semi-quantitative and quantitative methods such as matrix-based evaluations, risk assessments, and indicator-based analyses become feasible (Chebaeva et al., 2021).

3 Methods

To identify capability needs related to assessing product sustainability performance, a multiple case study involving four companies was conducted (Eisenhardt and Graebner, 2007), as shown in Table 1. A purposeful sampling strategy (Palinkas et al., 2015) was applied and the companies were selected to ensure variation in sector and because they have been working to include sustainability aspects in the PDP beyond compliance. The study focused on investigating their product development planning, as this activity links business strategy with product development and therefore plays a key role for efforts to implement SPD (Ulrich and Eppinger, 2012; Vilochani et al., 2025).

Table 1. Information about the case companies.

	Company			
	A	B	C	D
Sector	Aerospace	Food	Clean Tech	Construction Equipment
Size	Large	Medium	Small	Large
Workshop participants	7	3	2	7
Survey respondents	4	3	2	4
Interviewees	3	1	2	3

Data collection included workshops, a survey, and interviews. The workshops were facilitated by the research team to stimulate discussions about product strategy and PDP, as well as the decisions, stakeholders, and information flows characterizing each stage. A follow-up survey and interviews were conducted to gain in-depth insights into the knowledge gaps that impact the implementation of SPD. Roles involved in the different data collection activities included sustainability specialists, business developers, purchasing managers, product managers, risk and R&D specialists, as well as managing directors. These roles were chosen to collectively ensure coverage of expertise across different stages of the PDP, as well as experience with the broader context of the company's corporate and sustainability strategies.

All data underwent qualitative content analysis based on inductive thematic coding (Corbin and Strauss, 2008; Flick, 2007), which was consolidated into case narratives from which common themes, i.e., shared capability needs, were identified across the cases (Maxwell and Miller, 2008). Following the identification of empirical themes, an abductive reasoning logic was applied to synthesize these themes into higher-level capability needs, iteratively relating observed patterns to new conceptual capability categories (Dorst, 2011).

4 Findings

4.1 Capability needs related to assessing product sustainability performance

All four case companies aim at implementing SPD and consider the practice of assessing product sustainability performance as a critical element. Four main capability needs related to SPD implementation emerged from the data analysis. The identified capability needs are closely related to supporting the integration of sustainability considerations in decision-making about new product development, as well as supporting communication of sustainability efforts and benefits to customers. In this context, assessing product sustainability performance is a key element for fulfilling these capability needs. Product sustainability performance assessments were also seen as a practice that is present in current and emergent regulations that will likely impact and gain relevance for some companies and their value chains. Although this relevance is acknowledged, there is a lack of systematic practice of assessing product sustainability performance, and the companies highlighted their challenges and needs during data collection.

Capability 1 (C1) - Translating sustainability into business value: When seeking to integrate sustainability considerations into their PDP and decision-making, the companies expressed challenges in demonstrating how such integration could generate business value. This challenge was often related to uncertainties regarding future market trends and regulatory scenarios, particularly in relation to sustainability. In general, sustainability considerations were integrated when clearly stated as a customer requirement. Although three companies demonstrated a willingness to go beyond this, they struggled to articulate convincing arguments for the value of sustainability integration in product development in

relation to different risks, opportunities, market and regulatory scenarios. Participants mentioned that results from product sustainability performance assessments could provide information to support and improve communication of sustainability value for internal stakeholders, to support strategic decision-making, as well as to customers, supporting the creation of a sustainability business case.

Capability 2 (C2) - Aligning sustainability KPIs with corporate strategy and financial metrics: Closely linked to the need to translate sustainability considerations into business value, a second capability need relates to identifying how product sustainability performance indicators connected to and impacted other strategic and financial metrics at the corporate level. Participants from one company reported that their corporate sustainability goals were influenced by the product development roadmap; however, there was a lack of connection between the two that enabled assessing this impact and making decisions with the broader strategy in mind. The same challenge was highlighted regarding corporate long-term strategic planning. Participants from three companies expressed a specific need to measure the impact of sustainability on return on investment (ROI). For example, connecting the integration of sustainability considerations and ROI could enable the claim for a higher budget for research and development.

Capability 3 (C3) - Selecting sustainability criteria and KPIs: Current product sustainability performance assessment indicators used by the companies are mainly related to CO_2 emissions, as mentioned by participants from three companies. Emissions were seen as an aspect with higher data availability and easier communication with customers. In general, participants reported a lack of criteria and indicators related to social impacts. Participants from all four companies highlighted a lack of guidance on how to identify and select product sustainability performance indicators to support assessment, evaluation, and decision-making. This lack of guidance could also lead to uncertainty regarding what type of data companies should collect and use in different stages of the PDP.

Capability 4 (C4) - Weighting criteria and managing trade-offs: When participants discussed including sustainability criteria in product development decision-making, a common theme concerned how to assess the weight of sustainability aspects in relation to traditional design and business criteria, such as cost, profit margins, quality, and more. After deciding on the weighting of different criteria, trade-offs could arise both between the different types of criteria and when comparing the sustainability performance of different alternatives in relation to the specific sustainability criteria. Participants highlighted that in some cases regulatory requirements also created such trade-offs and that managing trade-offs requires KPIs and concrete data. For example, one company had guidance on how to weight sustainability criteria; however, if data were not available for a given criterion, it would not be included in the final weighting.

The four identified capability needs were frequently discussed in relation to one another during the interviews, indicating that companies perceive them as interdependent. Building on these empirical observations, an analytical synthesis of the findings suggests that demonstrating the business value of sustainability (C1) relies on the availability of clear sustainability criteria and indicators (C3) and their alignment with corporate strategy (C2). In turn, this alignment provides a basis for managing trade-offs between sustainability and traditional design or business criteria (C4). The relationship between these empirically identified capability needs and the task of assessing product sustainability performance across different phases of the PDP is further explored in the following section.

4.2 Conceptual model for product sustainability performance assessments

Based on the identified capability needs and literature on PDP, a conceptual model for product sustainability performance assessments was developed. The model, shown in Figure 1, shows how the task of assessing product sustainability performance evolves throughout the PDP, unfolding as an iterative sequence of activities that support different types of decisions. Across stages, these activities vary in purpose and are associated with shifts in the performance lens used to interpret product sustainability performance assessments (from effectiveness to efficiency). These differences in purpose and lens inform the required characteristics of assessment supports, providing a foundation to structure coherent methodological support.

Rather than viewing the task of assessing product sustainability performance as a single or static task, the model conceptualizes it as a sequence of interconnected activities that respond to different needs at different PDP stages. The outcomes of each activity inform subsequent assessments and decisions while also contributing to continuous learning and feedback, reinforcing the iterative nature of the PDP. To identify the main assessment activities, the analysis builds on established product development stages and tasks and their connections to SPD practices that critically influence product sustainability performance, as summarized in section 2. Based on this synthesis, specific product sustainability performance assessment activities and their descriptions are derived and connected to the identified capability needs, as presented in Table 2.

Figure 1. Conceptual model for product sustainability performance assessment.

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Table 2. Summary of product sustainability performance assessment activities.

Product sustainability performance assessment activity	Description	Connection to capability needs	References
Integrate sustainability into the product development strategy	This activity concerns establishing sustainability as an explicit consideration in product development planning, ensuring that sustainability ambitions are reflected in strategic decisions about product portfolios and development priorities.	C1	Diaz et al., 2021; Schulte et al., 2020; Vilochani et al., 2025
Link product-related sustainability goals into the product strategy	This task involves translating corporate or business-level sustainability objectives into concrete product-related sustainability goals that guide development decisions and investment choices.	C2	Nappi et al., 2024; Vilochani et al., 2025
Define product sustainability criteria and indicators	This task focuses on identifying relevant sustainability criteria and corresponding performance indicators that enable structured assessment, evaluation, and communication of product sustainability performance.	C2, C3	Issa et al., 2015; Kravchenko et al., 2019; Vilochani et al., 2025; Watz and Hallstedt, 2024
Assess concepts sustainability performance	This task entails exploring and qualitatively or semi-quantitatively evaluating the sustainability performance of early design concepts under conditions of high uncertainty and limited data availability.	All	Chebaeva et al., 2021; Hallstedt, 2017; Schöggl et al., 2017
Assess detailed design sustainability performance	This task involves quantitatively evaluating the sustainability performance of defined product solutions, supporting comparison, validation, and optimization of design alternatives.	All	Chebaeva et al., 2021; Peace et al., 2018
Manage trade-offs within sustainability criteria and across traditional criteria	This task addresses the evaluation and balancing of trade-offs both among sustainability criteria and between sustainability and traditional design or business criteria in order to support informed decision-making.	C4	Parolin et al., 2025; Vilochani et al., 2025

Assessing product sustainability performance initially needs to focus on clarifying what constitutes sustainability performance and, consequently, what the product should achieve or embody from a sustainability perspective (Watz and Hallstedt, 2024). At this stage, the concept of product sustainability performance plays a normative role (Sala et al., 2015), guiding companies in defining desired directions and priorities rather than measuring impacts. Companies translate this understanding into their specific context by connecting it to their business, sustainability, and product strategies. This typically occurs during product planning, when companies analyze strategic and contextual factors, identify opportunities, and select product development projects to pursue (Krishnan and Ulrich, 2001).

The first activity is to integrate a sustainability perspective into the product development strategy. Product sustainability performance assessment at this stage should provide an overview of the current situation, as well as future risks and opportunities, helping identify the most relevant sustainability aspects to address (Watz and Hallstedt, 2024). This step allows the application of an effectiveness performance lens, ensuring that decisions about new projects and product development focus on the most relevant sustainability aspects. The purpose is therefore not to measure impacts or indicators, but to define material sustainability aspects that are the most relevant to assess moving forward into product design (Watz and Hallstedt, 2022). This materiality analysis should be conducted from a systemic perspective, considering both the company and the product within the socio-ecological system (Zimek and Baumgartner, 2024), which requires the involvement of diverse stakeholders and close collaboration between business and product functions.

To support the integration of sustainability in product strategy and decision-making, it is important, after deciding on the material aspects, to map connections between product sustainability performance indicators and corporate KPIs and goals. Such alignment strengthens the influence of sustainability considerations in decision-making, as identified in both the empirical findings and the literature (Nappi et al., 2024; Vilochani et al., 2025), by clarifying how SPD can support value creation at the business-level. At this stage, the company should have a clear picture of what the product should do in terms of sustainability.

These two previous activities lead to the definition of product sustainability criteria and indicators, as companies move from a broader product strategy to a specific product development project. Sustainability aspects identified at the strategic level are translated into concrete criteria and measurable indicators that can be used in concept development and product realization, supporting assessments both in early and detailed design (Hallstedt, 2017; Nappi et al., 2024; Watz and Hallstedt, 2018). These indicators will be used as the basis for the assessment of, first, concept alternatives, and later detailed designs. The assessments now aim to inform how well the design alternatives perform relative to their intended objectives, which were defined in the earlier tasks. The differences in assessing concepts or detailed designs are mostly related to the level of design freedom and the decisions that have already been taken, what will affect the type of data available and impact, for example, if a qualitative or quantitative assessment is more applicable (Chebaeva et al., 2021). Additionally, when assessing early concepts, there is a tendency to have a stronger focus on effectiveness to ensure that the right decisions are being made. Moving into detailed design, an efficiency lens starts to be predominant, seeking to support decisions that lead to design optimization.

This cascading of material aspects to criteria and indicators increases the potential for sustainability priorities defined at higher levels to be carried into design decisions (Diaz et al., 2021; Nappi et al., 2024; Watz and Hallstedt, 2018). This could enable an iterative process in which sustainability aspects are assessed from both an effectiveness and efficiency performance lens, since there is coherence between aspects defined at the strategic level with indicators that are assessed during concept development and product realization. Therefore, results that inform how well a product is doing (efficiency), can be evaluated in relation to what it is supposed to do (effectiveness). This could be of critical importance in ensuring that decision-making in early and/or detailed concept design are not taken solely based on an efficiency perspective.

When making decisions throughout the process, questions regarding how to weight different criteria and manage trade-offs inevitably emerge. At this stage, sustainability performance must be considered alongside traditional design and business criteria, such as cost, technical feasibility, risk, and customer requirements. Improvements in one dimension may therefore lead to compromises in others, and different stakeholders may prioritize criteria differently (Parolin et al., 2025). The task thus shifts from evaluating single indicators to balancing multiple and sometimes conflicting objectives. The assessment at this stage provides structured information to make criteria, assumptions, and priorities explicit, enabling transparent comparison of alternatives and supporting dialogue among stakeholders. From an effectiveness perspective, trade-off management supports clarifying which sustainability aspects should be prioritized and why, particularly in early stages characterized by high uncertainty and qualitative judgments (Watz and Hallstedt, 2022). From an efficiency perspective, it supports comparing how well alternative solutions perform against defined criteria, enabling more quantitative balancing and optimization of competing objectives (Chebaeva et al., 2021). Approaches such as multi-criteria and decision analysis can combine quantitative indicators with qualitative judgments, clarify the implications of different weighting choices, and explore the robustness of decisions under uncertainty (Chebaeva et al., 2021; Parolin et al., 2025).

5 Concluding discussion

This study contributes to ongoing research on coherent methodological support for assessing product sustainability performance by showing how product development tasks and practices across the PDP relate to companies' capability needs and shape the types of support required at different stages. Findings reinforce that assessing product sustainability performance is not a single task in the PDP or just a matter of choosing the best tool – it is a set of activities that support different decisions at different stages. In early stages, product sustainability performance assessment has primarily a strategic role, supporting the company in understanding what sustainability means for product development (Peace et al., 2018; Vilochani et al., 2025; Watz and Hallstedt, 2024). Ideally, this is done from a socio-ecological systems and a business perspective, clarifying *what a product should do* in terms of sustainability (Watz and Hallstedt, 2018) and applying an effectiveness performance lens. As development progresses and product design becomes more concrete, assessment shifts toward evaluating and comparing alternatives to understand *how well the product does what it is supposed to do*, aiming to support decisions that lead to optimal choices (Chebaeva et al., 2021; Vilochani et al., 2025) from an efficiency point of view. This shift suggests that the main challenge is not the availability of methods, but how assessment activities are structured and connected so they support this change of perspectives and needs thorough the PDP. From a conceptual perspective, this distinction between effectiveness and efficiency helps explain why different types of product sustainability performance assessment approaches are needed at different stages. Early decisions require methods that

support reflection, prioritization, and alignment with strategic goals (Nappi et al., 2024; Vilochani et al., 2025), whereas later decisions benefit from quantitative comparisons and optimization (Chebaeva et al., 2021).

Rather than treating product sustainability performance assessments as isolated exercises, this study's findings suggest that they should be organized as a coherent sequence in which early strategic product sustainability performance assessments unfold into criteria and indicators that subsequently guide concept evaluation and trade-off analysis. Therefore, companies and practitioners can benefit from treating product sustainability performance assessment as a sequence of iterative activities within the PDP. In this sense, the study contributes to ongoing discussions on coherent methodological support (Diaz et al., 2021; Faludi, 2017; Gericke et al., 2020; Peace et al., 2018) by clarifying how assessment activities can be structured and connected across the PDP. To support such sequencing, companies need clear guidance on when and how to select, position, and combine different product sustainability performance assessment approaches throughout the PDP. The conceptual model proposed in this study may serve as such a structure by linking assessment purpose, performance lens (effectiveness and efficiency), decision context, and the input and output flows from one activity to another. Furthermore, moving toward coherent methodological support implies developing approaches that are designed to work in combination. This includes, for example, approaches that are explicit about their purposes, applicability to certain PDP stages, and their required inputs and expected outputs, as suggested in the notion of a methods ecosystem (Gericke et al., 2020). The conceptual model proposed in this study may serve as a structure to build coherent methodological support by showing how product sustainability performance assessment activities shift from effectiveness-oriented framing at strategic levels to efficiency-oriented evaluation at concept and detailed design stages, and by showing how results from one activity could cascade and propagate to the following stages in the PDP.

While this study contributes by synthesizing how the task of assessing product sustainability performance unfolds in the PDP and how it connects to empirically identified capability needs, it is important to acknowledge its limitations. The identified capability needs are based on a multiple case study involving four companies and reflect interpretations derived from qualitative data. Although purposeful sampling ensured variation in sector and size, the findings do not represent an exhaustive set of organizational needs and may not capture all contextual variations. The proposed model is conceptual and while it provides insights that could be valuable to support the development of coherent methodological support, it has not been empirically tested to explore how it could guide practitioners when choosing product sustainability performance assessment approaches, nor how researchers could use it to develop or improve assessment approaches. Finally, it is important to recognize that the implementation of product sustainability performance assessments depends on, and is interconnected with, other management and support practices of SPD, e.g. supply chain management, data management, sustainable manufacturing, communication (Vilochani et al., 2025). The proposed model should therefore be understood as one component within a broader range of practices required for effective SPD implementation. Verification of the proposed model is planned through additional case studies across varied industrial contexts, aimed at testing whether it adequately captures how assessment activities unfold throughout the PDP. This could be complemented by participatory engagements with practitioners, in which the model is used to guide the selection and sequencing of assessment approaches, allowing its usefulness as methodological support to be evaluated and the model iteratively refined. Such work will also examine how individual assessment approaches can be developed and adapted to function as part of a coherent methodological support.

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References

Ahmad, S., Wong, K.Y., Tseng, M.L., Wong, W.P., 2018. Sustainable product design and development: A review of tools, applications and research prospects. Resour. Conserv. Recycl. 132, 49–61. https://doi.org/10.1016/j.resconrec.2018.01.020

Albers, A., Reinemann, J., Hirschter, T., Fahl, J., Heitger, N., 2019. Validation-Driven Design in the Early Phase of Product Development. Procedia CIRP 84, 630–637. https://doi.org/10.1016/j.procir.2019.04.211

Bengtsson, M., Alfredsson, E., Cohen, M., Lorek, S., Schroeder, P., 2018. Transforming systems of consumption and production for achieving the sustainable development goals: moving beyond efficiency. Sustain. Sci. 13, 1533–1547. https://doi.org/10.1007/s11625-018-0582-1

Bjørn, A., Chandrakumar, C., Boulay, A.-M., Doka, G., Fang, K., Gondran, N., Hauschild, M.Z., Kerkhof, A., King, H., Margni, M., McLaren, S., Mueller, C., Owsianiak, M., Peters, G., Roos, S., Sala, S., Sandin, G., Sim, S., Vargas-Gonzalez, M., Ryberg, M., 2020. Review of life-cycle based methods for absolute environmental sustainability assessment and their applications. Environ. Res. Lett. 15, 083001. https://doi.org/10.1088/1748-9326/ab89d7

Broman, G.I., Robèrt, K.-H., 2017. A framework for strategic sustainable development. J. Clean. Prod. 140, 17–31. https://doi.org/10.1016/j.jclepro.2015.10.121

Brones, F.A., Carvalho, M.M.D., Zancul, E.D.S., 2017. Reviews, action and learning on change management for ecodesign transition. J. Clean. Prod. 142, 8–22. https://doi.org/10.1016/j.jclepro.2016.09.009

Buijs, J., 2003. Modelling Product Innovation Processes, from Linear Logic to Circular Chaos. Creat. Innov. Manag. 12, 76–93. https://doi.org/10.1111/1467-8691.00271
Ceschin, F., Gaziulusoy, I., 2016. Evolution of design for sustainability: From product design to design for system innovations and transitions. Des. Stud. 47, 118–163. https://doi.org/10.1016/j.destud.2016.09.002
Chebaeva, N., Lettner, M., Wenger, J., Schöggl, J.-P., Hesser, F., Holzer, D., Stern, T., 2021. Dealing with the eco-design paradox in research and development projects: The concept of sustainability assessment levels. J. Clean. Prod. 281, 125232. https://doi.org/10.1016/j.jclepro.2020.125232
Corbin, J., Strauss, A., 2008. Basics of Qualitative Research (3rd ed.): Techniques and Procedures for Developing Grounded Theory. SAGE Publications, Inc., 2455 Teller Road, Thousand Oaks California 91320 United States. https://doi.org/10.4135/9781452230153
Diaz, A., Schöggl, J.-P., Reyes, T., Baumgartner, R.J., 2021. Sustainable product development in a circular economy: Implications for products, actors, decision-making support and lifecycle information management. Sustain. Prod. Consum. 26, 1031–1045. https://doi.org/10.1016/j.spc.2020.12.044
Dorst, K., 2011. The core of 'design thinking' and its application. Des. Stud. 32, 521–532. https://doi.org/10.1016/j.destud.2011.07.006
Eisenhardt, K.M., Graebner, M.E., 2007. Theory Building From Cases: Opportunities And Challenges. Acad. Manage. J. 50, 25–32. https://doi.org/10.5465/amj.2007.24160888
Faludi, J., 2017. Recommending sustainable design practices by characterising activities and mindsets. Int. J. Sustain. Des. 3, 100. https://doi.org/10.1504/IJSDES.2017.091728
Faludi, J., Hoffenson, S., Kwok, S.Y., Saidani, M., Hallstedt, S.I., Telenko, C., Martinez, V., 2020. A Research Roadmap for Sustainable Design Methods and Tools. Sustainability 12, 8174. https://doi.org/10.3390/su12198174
Flick, U., 2007. Designing Qualitative Research. SAGE Publications, Ltd, 1 Oliver's Yard, 55 City Road, London England EC1Y 1SP United Kingdom. https://doi.org/10.4135/9781849208826
Gaziulusoy, A.I., 2015. A critical review of approaches available for design and innovation teams through the perspective of sustainability science and system innovation theories. J. Clean. Prod. 107, 366–377. https://doi.org/10.1016/j.jclepro.2015.01.012
Gaziulusoy, A.I., Brezet, H., 2015. Design for system innovations and transitions: a conceptual framework integrating insights from sustainablity science and theories of system innovations and transitions. J. Clean. Prod. 108, 558–568. https://doi.org/10.1016/j.jclepro.2015.06.066
Gebler, M., Juraschek, M., Thiede, S., Cerdas, F., Herrmann, C., 2022. Defining the "Positive Impact" of socio-technical systems for absolute sustainability: a literature review based on the identification of system design principles and management functions. Sustain. Sci. 17, 2597–2613. https://doi.org/10.1007/s11625-022-01168-1
Gericke, K., Eckert, C., Campean, F., Clarkson, P.J., Flening, E., Isaksson, O., Kipouros, T., Kokkolaras, M., Köhler, C., Panarotto, M., Wilmsen, M., 2020. Supporting designers: moving from method menagerie to method ecosystem. Des. Sci. 6, e21. https://doi.org/10.1017/dsj.2020.21
Hallstedt, S.I., 2017. Sustainability criteria and sustainability compliance index for decision support in product development. J. Clean. Prod. 140, 251–266. https://doi.org/10.1016/j.jclepro.2015.06.068
Hallstedt, S.I., Isaksson, O., 2017. Material criticality assessment in early phases of sustainable product development. J. Clean. Prod. 161, 40–52. https://doi.org/10.1016/j.jclepro.2017.05.085
Hauschild, M.Z., Kara, S., Røpke, I., 2020. Absolute sustainability: Challenges to life cycle engineering. CIRP Ann. 69, 533–553. https://doi.org/10.1016/j.cirp.2020.05.004
Kalish, D., Burek, S., Costello, A., Schwartz, L., Taylor, J., 2018. Integrating Sustainability into New Product Development: Available tools and frameworks can help companies ensure that sustainability is embedded as a fundamental building block of new product development. Res.-Technol. Manag. 61, 37–46. https://doi.org/10.1080/08956308.2018.1421379
Khurana, A., Rosenthal, S.R., 1998. Towards Holistic "Front Ends" In New Product Development. J. Prod. Innov. Manag. 15, 57–74. https://doi.org/10.1111/1540-5885.1510057
Kishita, Y., Höjer, M., Quist, J., 2024. Consolidating backcasting: A design framework towards a users' guide. Technol. Forecast. Soc. Change 202, 123285. https://doi.org/10.1016/j.techfore.2024.123285
Kravchenko, M., Pigosso, D.Ca., McAloone, T.C., 2019. Towards the ex-ante sustainability screening of circular economy initiatives in manufacturing companies: Consolidation of leading sustainability-related performance indicators. J. Clean. Prod. 241, 118318. https://doi.org/10.1016/j.jclepro.2019.118318
Krishnan, V., Ulrich, K.T., 2001. Product Development Decisions: A Review of the Literature. Manag. Sci. 47, 1–21. https://doi.org/10.1287/mnsc.47.1.1.10668
López-Mesa, B., Bylund, N., 2011. A study of the use of concept selection methods from inside a company. Res. Eng. Des. 22, 7–27. https://doi.org/10.1007/s00163-010-0093-2
Lovdahl, J., Schulte, J., I. Hallstedt, S., 2024. A Literature Review of Approaches for Assessing Product Sustainability Performance in Early Phases of the Product Innovation Process, in: Proceedings of NordDesign 2024. Presented at the NordDesign 2024, Reykjavik, Iceland, 12th - 14th August 2024, The Design Society, pp. 730–740. https://doi.org/10.35199/NORDDESIGN2024.78
Mallalieu, A., Isaksson Hallstedt, S., Isaksson, O., Watz, M., Almefelt, L., 2024. Barriers and enablers for the adoption of sustainable design practices using new design methods – Accelerating the sustainability transformation in the manufacturing industry. Sustain. Prod. Consum. 51, 137–158. https://doi.org/10.1016/j.spc.2024.08.023
Maxwell, J.A., Miller, B.A., 2008. Categorizing and connecting strategies in qualitative data anaylsis, in: Handbook of Emergent Methods. Guilford Press, New York, USA.
McAloone, T.C., Tan, A.R., 2005. Sustainable product development through a life-cycle approach to product and service creation: An exploration of the extended responsibilities and possibilities for product developers. Presented at the Proceedings of Eco-X Conference: Ecology and Economy in Electronix (Vol. CD-ROM, pp. 1-12). KERP, Austria.
Mendoza, J.M.F., Sharmina, M., Gallego-Schmid, A., Heyes, G., Azapagic, A., 2017. Integrating Backcasting and Eco-Design for the Circular Economy: The BECE Framework. J. Ind. Ecol. 21, 526–544. https://doi.org/10.1111/jiec.12590

Nappi, V., Sousa-Zomer, T.T., Cauchick-Miguel, P.A., Rozenfeld, H., 2024. Developing an action-oriented performance framework for sustainability measurement in the new product development. Int. J. Product. Perform. Manag. 73, 3133–3156. https://doi.org/10.1108/IJPPM-06-2023-0300

Palinkas, L.A., Horwitz, S.M., Green, C.A., Wisdom, J.P., Duan, N., Hoagwood, K., 2015. Purposeful Sampling for Qualitative Data Collection and Analysis in Mixed Method Implementation Research. Adm. Policy Ment. Health Ment. Health Serv. Res. 42, 533–544. https://doi.org/10.1007/s10488-013-0528-y

Parolin, G., Arnbjerg, J., Eriksen, H.A., McAloone, T.C., Pigosso, D.C.A., 2024. Enabling environmental sustainability and circularity assessment in technology development: The Value-Impact Scanner. Sustain. Prod. Consum. 49, 92–103. https://doi.org/10.1016/j.spc.2024.06.015

Parolin, G., Arnbjerg, J., McAloone, T.C., Pigosso, D.C.A., 2025. Navigating trade-offs in design for sustainability and circularity: A tool for decision making leveraging narratives and multiple criteria decision analysis. J. Clean. Prod. 494, 145004. https://doi.org/10.1016/j.jclepro.2025.145004

Peace, A., Ramirez, A., Broeren, M.L.M., Coleman, N., Chaput, I., Rydberg, T., Sauvion, G.-N., 2018. Everyday Industry—Pragmatic approaches for integrating sustainability into industry decision making. Sustain. Prod. Consum. 13, 93–101. https://doi.org/10.1016/j.spc.2017.08.003

Phaal, R., Farrukh, C.J.P., Probert, D.R., 2004. Technology roadmapping—A planning framework for evolution and revolution. Technol. Forecast. Soc. Change 71, 5–26. https://doi.org/10.1016/S0040-1625(03)00072-6

Rotondo, B., Bakker, C., Balkenende, R., Arquilla, V., 2025. Integrating Circular Economy Principles in the New Product Development Process: A Systematic Literature Review and Classification of Available Circular Design Tools. Sustainability 17, 4155. https://doi.org/10.3390/su17094155

Sala, S., Ciuffo, B., Nijkamp, P., 2015. A systemic framework for sustainability assessment. Ecol. Econ. 119, 314–325. https://doi.org/10.1016/j.ecolecon.2015.09.015

Schöggl, J.-P., Baumgartner, R.J., Hofer, D., 2017. Improving sustainability performance in early phases of product design: A checklist for sustainable product development tested in the automotive industry. J. Clean. Prod. 140, 1602–1617. https://doi.org/10.1016/j.jclepro.2016.09.195

Schulte, J., Villamil, C., Hallstedt, S.I., 2020. Strategic Sustainability Risk Management in Product Development Companies: Key Aspects and Conceptual Approach. Sustainability 12, 10531. https://doi.org/10.3390/su122410531

Ulrich, K.T., Eppinger, S.D., 2012. Product design and development, 5. ed., internat. ed. ed. McGraw-Hill, New York.

Villamil, C., Hallstedt, S., 2021. Sustainabilty integration in product portfolio for sustainable development: Findings from the industry. Bus. Strategy Environ. 30, 388–403. https://doi.org/10.1002/bse.2627

Villamil, C., Schulte, J., Hallstedt, S., 2022. Sustainability risk and portfolio management—A strategic scenario method for sustainable product development. Bus. Strategy Environ. 31, 1042–1057. https://doi.org/10.1002/bse.2934

Vilochani, S., McAloone, T.C., Pigosso, D.C.A., 2025. Bridging the gap: Exploring the effect of sustainable product development practices into product sustainability performance in manufacturing companies. J. Environ. Manage. 378, 124821. https://doi.org/10.1016/j.jenvman.2025.124821

Watz, M., Hallstedt, S.I., 2024. Depth and Detail or Quick and Easy? Benefits and Drawbacks of Two Approaches to Define Sustainability Criteria in Product Development, in: Fukushige, S., Kobayashi, H., Yamasue, E., Hara, K. (Eds.), EcoDesign for Sustainable Products, Services and Social Systems I. Springer Nature Singapore, Singapore, pp. 413–427. https://doi.org/10.1007/978-981-99-3818-6_28

Watz, M., Hallstedt, S.I., 2022. Towards sustainable product development – Insights from testing and evaluating a profile model for management of sustainability integration into design requirements. J. Clean. Prod. 346, 131000. https://doi.org/10.1016/j.jclepro.2022.131000

Watz, M., Hallstedt, S.I., 2018. Addressing Sustainability in Product Requirements from a Systems Perspective, in: DS 91: Proceedings of NordDesign 2018, Linköping, Sweden, 14th - 17th August 2018, NordDESIGN.

Zimek, M., Baumgartner, R.J., 2024. Systemic sustainability assessment: Analyzing environmental and social impacts of actions on sustainable development. Clean. Prod. Lett. 7, 100064. https://doi.org/10.1016/j.clpl.2024.100064